RESEARCH STATEMENT: EXTENDING COMPLEX NON-ABELIAN HODGE THEORY TO MODULI STACKS

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1. Background

The context. The complex non-Abelian Hodge (NAH) correspondence [Hit87; Cor88; Sim88; Sim92] is an equivalence between the differential graded category of semistable Higgs bundles and that of flat bundles over a compact Kähler manifold. This equivalence restricts to the full subcategories of polystable (stable) Higgs bundles and semisimple (simple/irreducible) flat connections. The induced bijection of coarse moduli spaces restricts to a homeomorphism between the polystable locus on the the Dolbeault side and the semisimple locus on the de Rham side [Sim94, Theorem 7.18], but is known to have discontinuities on the semistable, non-polystable locus [Sim94, Counterexample on pages 38 — 39]. This has raised interest in finding a stack-level NAH correspondence [Sim96, p. 12]. The Borel-Moore homologies of the respective moduli stacks have recently been proved to be isomorphic in the cases of curves of genus 0 and 1 by Ben Davison [Dav23] and those of arbitrary genus by Lucien Hennecart [Hen25]. In our most recent pair of articles [AR26a; AR26b], my doctoral supervisor and I have partially answered the question of an equivalence between the moduli stacks themselves, or substacks thereof, over arbitrary compact Kähler manifolds. Several questions in this area, however, remain open and I wish to address some of these in my postdoctoral research.

Progress so far. In [AR26a], we constructed diffeologically presentable (in the sense of [RV18]) moduli stacks $\mathcal{M}_{\mathrm{Dol}}(X)$ and $\mathcal{M}_{\mathrm{dR}}(X)$, over the site of smooth manifolds, parametrizing smooth families of Higgs bundles and of flat bundles respectively over a fixed compact Kähler manifold X. By relativising the extension completion construction of dg-categories from [Sim92], we produced an equivalence of a substack $\mathcal{M}^{\mathcal{H}}_{\mathrm{Dol}}(X)$ of $\mathcal{M}_{\mathrm{Dol}}(X)$ with a substack $\mathcal{M}^{\mathcal{H}}_{\mathrm{dR}}(X)$ of $\mathcal{M}_{\mathrm{dR}}(X)$, mediated by a prestack $\mathcal{M}_{\mathcal{H}}(X)$ parametrizing harmonic bundles. These substacks contain, as their fibre groupoids over the point, the groupoids of semistable Higgs bundles and all flat bundles respectively. In addition, $\mathcal{M}^{\mathcal{H}}_{\mathrm{Dol}}(X)$ contains the family of [Sim94, Counterexample on pages 38 — 39]. Therefore, our equivalence is an extension of the classical equivalence of categories to moduli stacks that also accounts for parts of the semistable, non-polystable locus where the bijection of coarse moduli spaces becomes discontinuous.

In [AR26b], we perform a detailed study of the infinitesimal properties of this equivalence and the loci it accounts for. We prove that $\mathcal{M}^{\mathcal{H}}_{\mathrm{Dol}}(X)$ contains the stable locus and provide examples showing that it does not contain the entire polystable locus, and hence, nor the semistable one. Nevertheless, identical methods produce the same results for larger moduli stacks $\mathcal{M}^{\mathcal{H}}_{\mathrm{Dol},d}(X) \subset \mathcal{M}_{\mathrm{Dol},d}(X)$ and $\mathcal{M}^{\mathcal{H}}_{\mathrm{dR},d}(X) \subset \mathcal{M}_{\mathrm{dR},d}(X)$ parametrizing analogous C^d families, giving extensions of the correspondence. However, Proposition 14.3 in the article gives an example of a real analytic family of polystable Higgs bundles that is outside $\mathcal{M}^{\mathcal{H}}_{\mathrm{Dol},d}(X)$ for all d. The reason this family is not in these stacks is that it admits no harmonic metric everywhere continuous in the parameter direction. Following this lead, we allowed harmonic metrics to be discontinuous in the parameter direction while still requiring Higgs fields and flat connections to be continuous. This resulted in an equivalence of strictly larger substacks $\mathcal{M}^{\mathcal{H}}_{\mathrm{Dol},0} \subset \mathcal{M}^{\mathrm{wk}\mathcal{H}}_{\mathrm{Dol},0}(X) \subset \mathcal{M}_{\mathrm{Dol},0}(X)$ and $\mathcal{M}^{\mathcal{H}}_{\mathrm{dR},0} \subset \mathcal{M}^{\mathrm{wk}\mathcal{H}}_{\mathrm{dR},0}(X) \subset \mathcal{M}_{\mathrm{dR},0}(X)$. The problematic family of Proposition 14.3 is in $\mathcal{M}^{\mathrm{wk}\mathcal{H}}_{\mathrm{Dol},0}$, and hence, we get a further extension of the NAH correspondence.

Proposed Research. The pattern seems to be that weakening the regularity conditions involved results in extensions of the stack-level NAH correspondence. However, it is not clear how far this phenomenon continues. At the same time, our work so far has been in the non-derived setting. It would be interesting to develop the derived version of this picture and try to achieve better coverage of the polystable locus. Finally, there might be alternative mechanisms to move between families of Higgs bundles and flat connections with better geometricity properties. I hope to address these matters in detail in my postdoctoral research. Please see this URL for a more concise description of this program.

2. Sporadic Harmonic Mediator

Consider the following example due to my doctoral supervisor:

Example 2.1. For an elliptic curve X, consider the complex vector bundle $E = (\pi_X^* \mathcal{O}_X)^{\oplus 3} \cong \pi_X^*\left(\mathcal{O}_X^{\oplus 3}\right)$ over $\mathbb{C} \times X$ obtained by pullback along the projection $\pi_X : \mathbb{C} \times X \longrightarrow X$. Let $\alpha \in H^0(X, \Omega_X^1)$ be a non-zero 1–form and:

$$\theta(z) = \begin{bmatrix} 0 & z & 0 \\ 0 & 0 & z \\ z^5 & 0 & 0 \end{bmatrix} \alpha$$

Then, (E, θ) is a holomorphic family of polystable Higgs bundles on X parametrized by $\mathbb{C}$.

We can show that for every choice of harmonic metric on each slice $(E_z, \theta(z))$ of the family (E, θ) over $\{z\} \times X, z \in \mathbb{C}$, even ones discontinuous in z, the family of operators $\{D_z\}_{z \in \mathbb{C}}$ obtained by applying the NAH transform on each $(E_z, \theta(z))$ have connection matrices approaching ∞ as z approaches 0, but the connection matrix of D_0 is 0. That is, the operator family $\{D_z\}_{z \in \mathbb{C}}$ is discontinuous at 0. This shows that even the stack $\mathcal{M}_{\mathrm{Dol},0}^{\mathrm{wk}\mathcal{H}}(X)$ does not contain the entire polystable locus. This also implies that any mapping from Higgs bundle families to flat bundle families that applies the NAH transform to each individual Higgs bundle cannot generate a continuous family of flat connections from Example 2.1, and hence cannot cover the entire polystable locus. We are thus forced to consider generalizations of the moduli stacks themselves. This is surprising as the NAH correspondence is a homeomorphism on the coarse polystable locus, and understanding the discrepancy in the case of Example 2.1 provides a lead for modifying the stack-level picture.

$(E_z, \theta(z))$ approaches the trivial Higgs bundle in the coarse moduli space of Higgs bundles as $z \longrightarrow 0$. On an open sector S of the punctured disk around $0 \in \mathbb{C}$, θ can be diagonalized as $\theta'(\lambda) = \mathrm{diag}(\lambda, \lambda\omega, \lambda\omega^2)$ where λ is a branch of $z^{7/3}$ and $\omega = e^{2\pi i/3}$, and its associated flat connection matrices become $C(\lambda) = \mathrm{diag}(\lambda + \bar{\lambda}, \lambda\omega + \bar{\lambda}\bar{\omega}, \lambda\omega^2 + \bar{\lambda}\bar{\omega}^2)$. These matrices both approach zero as z does. Hence, in the coarse moduli space of flat bundles, the family D_z approaches the trivial connection — the image of $(E_0, \theta(0))$ under the NAH transforma — as $z \longrightarrow 0$ on S. Now, let R be the ray in the punctured disk that is not in S. Choosing a different sector S' containing R and applying the same argument shows that D_z approaches the trivial connection, up to isomorphism, also along R. It is important to note that the slicewise isomorphisms of flat bundles do not assemble to a family of isomorphisms continuous at 0. We abstract the situation into the following definition:

Definition 2.2 (Sporadic Families)**.** A sporadic family of harmonic bundles consists of a tuple $(U, P, \{U_i\}_{i \in I}, E, \theta, \{D_i\}_{i \in I})$ where:

- U is a C^0 manifold.
- P is a closed subset of U.
- $\{U_i\}_{i \in I}$ is an open cover of $U \setminus P$.
- $E \longrightarrow U \times X$ is a complex vector bundle.

- (E, θ) is a continuous family of Higgs bundles on X parametrized by U.
- $(E|(U_i \times X), D_i)$ is a continuous family of flat bundles on X parametrized by U_i.
- For each $i \in I, u \in U_i$, $(E_u, \theta_u, D_{i,u})$ is a harmonic bundle.
- For each $i, j \in I, u \in U_i \cap U_j$, there is an isomorphism of flat bundles $(E_u, D_{i,u}) \cong (E_u, D_{j,u})$.
- For each $p \in P$, $(E|(\{p\} \times X), D_p)$ is a flat bundle on X.
- The $(E|(U_i \times X), D_i)$ extend to continuous families of flat bundles parametrized by $(P \cap \overline{U_i}) \cup U_i$, that are slicewise isomorphic on $P \cap \overline{U_i} \cap \overline{U_j}$.

Now, consider the portion $(U, P, \{U_i\}_{i \in I}, E, \{D_i\}_{i \in I})$ of a sporadic family of harmonic bundles, and remove the harmonicity requirement over the U_i. Then, we call such an object a sporadic family of flat bundles.

We should then be able to show pullback of such objects along C^0 maps $f : V \longrightarrow U$, and organize them into differential graded prestacks (or a presheaf of differential graded categories) over the site of C^0 manifolds. We should also be able to prove that sporadic families of flat bundles satisfy descent. The crux here is to show that the extension of local families of flat bundles to the distinguished closed subset in the parameter space interacts well with pullback and descent. Assuming we can do this, we make a definition:

Definition 2.3 (Sporadic (Pre)stacks)**.** Denote the dg-prestacks of sporadic harmonic bundle families, if they exists, as $\mathcal{M}_{\mathcal{H}}^{\text{sporadic}}(X)$. We will also call this the sporadic harmonic mediator prestack. Denote the stack of sporadic flat bundle families, if it exists, as $\mathcal{M}_{\text{dR}}^{\text{sporadic}}(X)$.

Note that our methods do not require harmonic mediators to satisfy descent and so we will not address this question for now. The natural questions that follow are:

Question 2.1. Is the stack $\mathcal{M}_{\text{dR}}^{\text{sporadic}}(X)$ diffeologically presentable?

Question 2.2. We should have forgetful maps $\mathcal{M}_{\text{Dol}}(X) \longleftarrow \mathcal{M}_{\mathcal{H}}^{\text{sporadic}}(X) \longrightarrow \mathcal{M}_{\text{dR}}^{\text{sporadic}}(X)$. Can the extension completion followed by stackification technique of [AR26a] be adapted to this sporadic setting?

If these questions can be answered positively, we have a further extension of the stack-level NAH correspondence accounting for Example 2.1 that is still geometric, noting that this comes at the cost of changing the definition of the de Rham side. The main question, then, would be:

Question 2.3. How much of the polystable locus in $\mathcal{M}_{\text{Dol}}(X)$ does the essential image of the stackified extension completion of $\mathcal{M}_{\mathcal{H}}^{\text{sporadic}}(X)$ cover? Does the regularity in the parameter directions of the isomorphisms of flat bundle families in the definition of sporadic families affect the coverage of the polystable locus?

3. Categorical and Derived Enhancements

Sporadic families of flat bundles as in Example 2.1 demonstrate the need for singular gluing of objects, and derived geometry likely has a role here that I would like to explore. In [AR26a], we defined dg-prestacks $\mathcal{M}_{\text{Dol}}(X), \mathcal{M}_{\text{dR}}(X)$ of Higgs bundles and flat bundles respectively whose object-wise homotopy categories give the stacks $\mathcal{M}_{\text{Dol}}(X), \mathcal{M}_{\text{dR}}(X)$ from the previous section. Instead of taking homotopy categories, if we take the categories of chain complexes with the usual model structures, we obtain corresponding derived moduli stacks. However, to get a fully derived picture, we should also consider derived parameter spaces in addition to smooth manifolds. This leads us to:

Question 3.1. Can smooth manifolds be replaced with derived C^∞–schemes in the constructions and arguments of [AR26a] to obtain similar notions of partial differentials, λ–connection families and ∞–categories of chain complexes thereof?

Definition 3.1 (Derived Moduli Stacks of Higgs Bundles and Connections)**.** If we can handle the previous question, we obtain derived prestacks parametrizing chain complexes of Higgs bundles, flat bundles and harmonic bundles which we denote by $\mathfrak{M}_{\mathrm{Dol}}(X)$, $\mathfrak{M}_{\mathrm{dR}}(X)$ and $\mathfrak{M}_{\mathcal{H}}(X)$. We will denote the sub-prestacks consisting of perfect complexes by $\mathfrak{M}^{\mathrm{perf}}_{\mathrm{Dol}}(X), \mathfrak{M}^{\mathrm{perf}}_{\mathrm{dR}}(X)$ and $\mathfrak{M}^{\mathrm{perf}}_{\mathcal{H}}(X)$.

$\mathfrak{M}_{\mathrm{Dol}}(X)$, $\mathfrak{M}_{\mathrm{dR}}(X)$ should be stacks but whether $\mathfrak{M}_{\mathcal{H}}(X)$ satisfies descent is less clear. Similar to the non-derived case, we will not address this question for now.

Question 3.2. Algebro-geometric versions of $\mathfrak{M}^{\mathrm{perf}}_{\mathrm{Dol}}(X), \mathfrak{M}^{\mathrm{perf}}_{\mathrm{dR}}(X)$ are known to be locally geometric (for example [PS20, Corollary 4.2.1]). Can we adapt the techniques used to obtain diffeological atlases in [AR26a] for constructing atlases in the derived setting?

We then have forgetful maps $\mathfrak{M}^{\mathrm{perf}}_{\mathrm{Dol}}(X) \longleftarrow \mathfrak{M}^{\mathrm{perf}}_{\mathcal{H}}(X) \longrightarrow \mathfrak{M}^{\mathrm{perf}}_{\mathrm{dR}}(X)$ leading to natural questions:

Question 3.3. Can we take extension completions and then stackifications of these maps to obtain an equivalence of substacks $\mathfrak{M}^{\mathrm{perf},\mathcal{H}}_{\mathrm{Dol}} \subset \mathfrak{M}^{\mathrm{perf}}_{\mathrm{Dol}}$ and $\mathfrak{M}^{\mathrm{perf},\mathcal{H}}_{\mathrm{dR}} \subset \mathfrak{M}^{\mathrm{perf}}_{\mathrm{dR}}$, as we did in the non-derived case? How much of the semistable and polystable loci in $\mathfrak{M}^{\mathrm{perf}}_{\mathrm{Dol}}(X)$ are covered by $\mathfrak{M}^{\mathrm{perf},\mathcal{H}}_{\mathrm{Dol}}$?

Remark 3.2. Note that since the notion of equivalence of objects is weaker than in the non-derived case, this has the potential to cover larger portions of the various loci.

Question 3.4. If we allow harmonic metrics to have discontinuities in the parameter, we should have a similar equivalence between substacks $\mathfrak{M}^{\mathrm{perf},\mathrm{wk}\mathcal{H}}_{\mathrm{Dol}}(X) \subset \mathfrak{M}^{\mathrm{perf}}_{\mathrm{Dol}}(X)$ and $\mathfrak{M}^{\mathrm{perf},\mathrm{wk}\mathcal{H}}_{\mathrm{dR}}(X) \subset \mathfrak{M}^{\mathrm{perf}}_{\mathrm{dR}}(X)$. Are these loci significantly bigger?

A further categorical generalization is possible. In [AR25], we constructed simplicial algebraic stacks $\mathcal{M}^{\bullet}_{\mathrm{Dol}}(X), \mathcal{M}^{\bullet}_{\mathrm{dR}}(X)$ where the n–th degree parametrizes diagrams of families of Higgs bundles and flat bundles respectively on a smooth complex projective variety X of shape Δ^n, the standard n–simplex or the nerve of the totally ordered set with $n+1$ elements. There are also derived algebraic geometric counterparts of these [PS23, §4]. Hence, we can expect there to be analogous simplicial derived C^∞–stacks $\mathfrak{M}^{\bullet}_{\mathrm{Dol}}(X), \mathfrak{M}^{\bullet}_{\mathrm{dR}}(X)$. These should be geometric in the sense of admitting atlases from derived C^∞–schemes levelwise. They should also form an ∞–category object internal to the ∞–category of derived C^∞–stacks, perhaps similar to the 2–Segality condition of [DPS25, Lemma II.4.1]. We should also have a simplicial derived (pre)stack $\mathfrak{M}^{\bullet}_{\mathcal{H}}(X)$ whose n–th degree parametrizes Δ^n–shaped diagrams of complexes of families of harmonic bundles. There should then be mappings of simplicial derived prestacks, over the site of C^∞–schemes, $\mathfrak{M}^{\bullet}_{\mathrm{Dol}}(X) \longleftarrow \mathfrak{M}^{\bullet}_{\mathcal{H}}(X) \longrightarrow \mathfrak{M}^{\bullet}_{\mathrm{dR}}(X)$, by similar arguments as in the non-derived case. The main questions would then be:

Question 3.5. Are these mappings, after a possible extension completion and then stackification, fully faithful in some appropriate sense suitable for simplicial objects in derived C^∞–stacks?

Question 3.6. If the answer to the previous question is yes, there should be an equivalence, again in the appropriate sense for simplicial derived C^∞–stacks, between simplicial substacks $\mathfrak{M}^{\mathcal{H},\bullet}_{\mathrm{Dol}}(X) \subset \mathfrak{M}^{\bullet}_{\mathrm{Dol}}(X)$ and $\mathfrak{M}^{\mathcal{H},\bullet}_{\mathrm{dR}}(X) \subset \mathfrak{M}^{\bullet}_{\mathrm{dR}}(X)$. How much of the loci of simplices with stable, polystable and semistable vertices do these cover?

Question 3.7. Does the coverage change significantly if we allow harmonic metric discontinuities in parameter directions?

Question 3.8. How do sporadic families of harmonic bundles and flat bundles fit into this setting of simplicial derived stacks?

4. Alternatives to the Correspondence

Our observations and results so far suggest that the correspondence between Higgs bundles and flat connections mediated by harmonic metrics requires more and more machinery to maintain its geometricity as we try to increase the coverage of the polystable or semistable loci on the Dolbeault side. The harmonicity condition is what forces our stack-level NAH correspondence to fail on parts of the polystable locus. Thus, we can consider the possibility of correcting its mechanism to account for all of the stack-level polystable locus. Our observation here is that the formula used to obtain a connection from a Higgs field given a metric might be one of many choices. Hence, we may ask:

Question 4.1. Can we modify the formula translating back and forth between a flat connection and a Higgs field using a metric, and then adapt the analysis of [Sim88] to produce these metrics with regularity in the parameter for all polystable families?

Question 4.2. Does this produce an equivalence of categories, moduli spaces, stacks as in usual non-Abelian Hodge theory?

It is also interesting to consider moving from Higgs bundles to flat bundles without mediation by harmonic metrics. The non-Abelian Hodge filtration stack provides motivation and guidance for this investigation. This is the stack whose objects are of the form (U, λ, E, D), where U is a smooth manifold, $\lambda : U \longrightarrow \mathbb{C}$ is a continuous function, $E \longrightarrow U \times X$ is a complex bundle and D is a λ–$d_{X/U}$–connection in the sense of [AR26a, §2.4]. Call such an object a Hodge family. Given a family of harmonic bundles (E, D'', D') and a harmonic metric family h continuous in the parameter, we have a standard Hodge family defined by:

$$\nabla_\lambda = \overline{\partial}_{E/U} + \partial_{E/U,h} + \lambda^{-1}\theta + \lambda\theta^{\dagger h}$$

where the operators on the right are the relative holomorphic structure, the Higgs field and their conjugates with respect to the metric, multiplied by λ or λ^{-1}. However, this is just one way to generate a Hodge family and it is natural to ask: given a family of Higgs bundles (E_0, D_0''), with constraints such as (poly/semi)stability if needed, is there another method to produce a Hodge family $\mathcal{F} := (U, \lambda, E, D)$ such that $\mathcal{F}_0 := (E, D)|\lambda^{-1}(0)$ is equivalent to (E_0, D_0'')?

Remark 4.1. Bertrand Eynard and Raphaël Belliard have discussed with me a method to produce such a Hodge family, using topological recursion, which is different from the standard Hodge family. They also added their method does not require many of the hypotheses of non-Abelian Hodge theory.

This is an exciting possibility and I hope to investigate the following questions in this context:

Question 4.3. Can we characterize $\mathcal{F}_1 := (E, D)|\lambda^{-1}(1)$ given a characterization of $\mathcal{F}_0$ such as (poly/semi)stability?

Question 4.4. How functorial is this construction of Hodge families? Can we organize it into a mapping of stacks $\mathcal{M}_{\mathrm{Dol}}(X) \longrightarrow \mathcal{M}_{\mathrm{Hodge}}(X)$ as the one from [AR26b, §13] given by the standard Hodge family construction?

Question 4.5. We have a mapping $\mathcal{F} \longmapsto \mathcal{F}_1$. Does this give a mapping of stacks $\mathcal{M}_{\mathrm{Hodge}}(X) \longrightarrow \mathcal{M}_{\mathrm{dR}}(X)$? If we can construct this chain of maps $\mathcal{M}_{\mathrm{Dol}}(X) \longrightarrow \mathcal{M}_{\mathrm{Hodge}}(X) \longrightarrow \mathcal{M}_{\mathrm{dR}}(X)$, is it an equivalence when restricted to a substantial locus of $\mathcal{M}_{\mathrm{Dol}}(X)$? Can we characterize such a locus? Can we characterize essential image in $\mathcal{M}_{\mathrm{dR}}(X)$?

Question 4.6. What is the derived enhancement of this picture? What is the enhancement in terms of simplicial derived stacks?

Question 4.7. Ultimately, does this increase the locus coverage compared to the NAH correspondence?

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